

Final Mathematical Verification Report

Reviewer: Pure mathematician (formal verification of mathematical claims only) **Date:** 2026-06-28 **Scope:** Theorems 1-5, supporting lemmas, and numerical test cases

Files reviewed: - G:\Xiaogan_Supercomputing_data\SCX\theory\theorems\

- G:\Xiaogan_Supercomputing_data\SCX\theory\theorems\02_weak_feature_failure.md

- G:\Xiaogan_Supercomputing_data\SCX\theory\theorems\03_unidentifiability_theorem.m

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\exact_constant_minimax.md

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\lemma_AB_bahadur_rao_f1.md

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\lemma_CD_chernoff_adaptive

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\lemma_EF_lowerbound_aggreg

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\cluster_consistency_v3.md

- G:\Xiaogan_Supercomputing_data\SCX\theory\explorations\deep_math_connections.md

1. Theorem 1 (Noise Detection Guarantee)

Lemma 1 (Mean Separation)

Claim: $E[C|\text{noise},x] = 1 - E[C|\text{clean},x]/(K-1)$

Verification: PASS.

The proof expands: $E[C|\text{noise},x] = (1/M) \sum_m P(f_m(x) \neq y \mid \text{noise}, x) - P(f_m(x) \neq y \mid \text{noise}, x) = \sum_{\{c \neq y\}} P(y=c/\text{noise}) P(f_m(x) \neq c \mid x) - \sum_{\{c \neq y\}} (1/(K-1)) (1 - P(f_m(x)=c|x)) - = 1 - (1/(K-1)) * \sum_{\{c \neq y\}} P(f_m(x)=c/x) - = 1 - (1/(K-1)) (1 - P(f_m(x)=y/x)) - = 1 - (1/(K-1)) E[e_m|\text{clean},x]$

Averaging over m gives: $E[C|\text{noise}, x] = 1 - E[C|\text{clean}, x]/(K-1)$. The algebra is correct. Each step uses only: (A4) uniform noise over $K-1$ classes, linearity of expectation, and identity $\sum_{c \neq y^*} P(f_m=c|x) = E[e_m|\text{clean}, x]$.

The separation condition $\mu_s < (K-1)/K$ is correctly derived from: clean mean $\leq \mu_s$, noise mean $\geq 1 - \mu_s/(K-1)$, requiring $\mu_s < 1 - \mu_s/(K-1) \Rightarrow \mu_s * K/(K-1) < 1 \Rightarrow \mu_s < (K-1)/K$.

The optimal threshold $\theta^* = (1/2)(1 + \mu_s^*(K-2)/(K-1))$ solves $\theta - \mu_s = 1 - \mu_s/(K-1) - \theta$ correctly.

Lemma 2 (FPR)

Claim: $P(C > \theta \mid \text{clean}, s) \leq \exp(-2M(\theta - \mu_s)^2)$

Verification: PASS.

Standard Hoeffding inequality for bounded $[0,1]$ independent r.v.s. The conditioning chain: - Fix x in s : $E[C|\text{clean}, x] \leq \mu_s$ by (A5) - $\{e_m\}$ conditionally independent given x by (A2), bounded by $[0,1]$ by (A3) - $P(C - E[C|x] > \theta - E[C|x] \mid \text{clean}, x) \leq \exp(-2M(\theta - E[C|x])^2) \leq \exp(-2M(\theta - \mu_s)^2)$ - The last step uses: $\theta > \mu_s \geq E[C|x]$, so $(\theta - E[C|x]) \geq (\theta - \mu_s)$ - Marginalizing over x in s preserves the bound via sup

All conditions for Hoeffding are satisfied.

Lemma 3 (TPR)

Claim: $P(C \leq \theta \mid \text{noise}, s) \leq \exp(-2M(1 - C_{\text{bal}}*\mu_s/(K-1) - \theta)^2)$

Verification: PASS, with careful reading.

The proof conditions on (x, c) where c is the noise label: - Given (x, c) , $\{e_m\}$ are conditionally independent (A1, A2) with $e_m \sim \text{Bern}(1 - \mu_{\{c,m\}}(x))$
 - $E[C|x,c] = 1 - \mu_c(x) \geq 1 - C_{\text{bal}}\mu_s/(K-1)$ by (A6) - The theorem assumes $\theta < 1 - C_{\text{bal}}\mu_s/(K-1)$, so Hoeffding applies - $P(C \leq \theta \mid x, c) \leq \exp(-2M(E[C|x,c] - \theta)^2) \leq \exp(-2M(1 - C_{\text{bal}}\mu_s/(K-1) - \theta)^2)$ - Averaging over c (uniform over $K-1$ classes) preserves the bound

Key subtlety: the bound uses $\mu_c(x) \leq C_{\text{bal}}\mu_s/(K-1)$, so $E[C|x,c] \geq 1 - C_{\text{bal}}\mu_s/(K-1)$. The Hoeffding gap $(E[C|x,c] - \theta)$ is minimized (worst case) when $E[C|x,c]$ is smallest, which occurs at the max $\mu_c(x)$. The A6 assumption correctly captures this via C_{bal} .

F1 bound

Claim: $F1 \geq 1 - (1/\eta) \sum_s \rho_s \exp(-2M \Delta_s^2)$

Verification: PASS.

Algebraic derivation: - $F1 = 2\eta \text{TPR} / (\eta(1+\text{TPR}) + (1-\eta)\text{FPR})$ [verified via Precision-Recall algebra] - Substituting $\text{TPR} \geq 1 - \delta_1$, $\text{FPR} \leq \delta_2$ gives denominator $\geq \eta - 1 - F1 \leq \delta_1 + (1-\eta)\delta_2/\eta$
 - Since $\delta_1, \delta_2 \leq \sum_s \rho_s \exp(-2M \Delta_s^2)$, the final bound follows

The denominator is strictly positive: $\eta(2-\text{FNR}) + (1-\eta)\text{FPR} \geq \eta > 0$ for $\eta > 0$.

2. Theorem 2 (Weak Features)

Fano inequality

Claim: $P(S_{\text{hat}} \neq S) \geq (H(S) - \delta - \log 2) / \log K$

Verification: PASS.

Standard Fano inequality: $P(S_{\text{hat}} \neq S) \geq (H(S|\phi(X)) - \log 2) / \log |S|$. Since $I(\phi; S) \leq \delta$ by the delta-weak feature definition: $H(S|\phi) = H(S) - I(\phi; S) \geq H(S) - \delta$. Substituting gives: $P(S_{\text{hat}} \neq S) \geq (H(S) - \delta - \log 2) / \log K$.

The feature-dependent Markov chain $Z \rightarrow S \rightarrow X \rightarrow \phi(X)$ correctly implies $I(\phi; Z) \leq I(S; Z) \leq H(Z)$ by the data processing inequality.

Pinsker inequality

Claim: $TV(P, P_{\text{tilde}}) \leq \sqrt{\delta/2}$

Verification: PASS.

$P_{\text{tilde}}(\phi, S) = P(\phi) * P(S)$. $KL(P \parallel P_{\text{tilde}}) = I(\phi; S) = \delta$. Pinsker: $TV \leq \sqrt{KL/2} = \sqrt{\delta/2}$.

F1 bound

Claim: $F1_{\text{SCX}} \leq F1_{\text{base}} + C_F * \sqrt{\delta/2}$

Verification: PASS (with caveat on Lipschitz constant).

The reasoning chain: 1. Under P_{tilde} , SCX detection score degrades to loss baseline: $NS(x) \sim \max_m D_m(x)$ 2. Hence $F1_{P_{\text{tilde}}}(\text{SCX}) = F1_{\text{base}}$ 3. $|F1_P(\text{SCX}) - F1_{P_{\text{tilde}}}(\text{SCX})| \leq C_F * TV(P_{\text{pred}}, P_{\text{tilde_pred}})$ by Lipschitz property 4. $TV(P_{\text{pred}}, P_{\text{tilde_pred}}) \leq TV(P, P_{\text{tilde}})$ by data processing inequality 5. $\leq \sqrt{\delta/2}$

The Lipschitz analysis of $F1(TP, FP, FN) = 2TP/(2TP+FP+FN)$ is partial derivatives and the document acknowledges that C_F depends on precision/recall regime. The specific values ($C_F \leq 2$ for precision, recall ≥ 0.1) are heuristic estimates, not rigorous bounds. This is flagged as an approximate constant in the text.

Minor issue: The AUC bound's TV-to-conditional-TV step uses division by η and $1-\eta$: $|P(A|Z=1) - P_{\text{tilde}}(A|Z=1)| \leq TV(P, P_{\text{tilde}})/\eta$. This requires $\eta > 0$ (non-zero noise rate). For $\eta \rightarrow 0$ the bound diverges, which the document correctly acknowledges.

3. Theorem 3 (Unidentifiability)

K=2 construction

Claim: $P_{\text{noise}}(x, y, \{f_m\}) = P_{\text{hard}}(x, y, \{f_m\})$

Verification: PASS.

World A (noise): s_1 has noise rate η , clean expert error ϵ_1 . s_2 has no noise. World B (hard): s_1 has random true label ($P(y=0)=1-\eta$, $P(y=1)=\eta$), experts biased toward class 0.

Equality verification: $P_A(y=0|s_1) = (1-\eta)1 + \eta 0 = 1-\eta$ -

$P_B(y=0|s1) = P(y=0|s1) = 1-\eta$ - $P_A(f_m=0|s1) = 1-\epsilonpsilon_1$
(expert accuracy against true label $y=0$) - $P_B(f_m=0|s1) = (1-\eta)(1-\epsilonpsilon_1) + \eta(1-\epsilonpsilon_1) = 1-\epsilonpsilon_1$

Joint factorization $P(x,y,\{f_m\}) = P(x) * P(y|x) * \prod_m P(f_m|x)$ holds by conditional independence in both constructions. State $s2$ is identical in both worlds.

K>2 construction

Claim: The construction works for all $K \geq 2$.

Verification: PASS.

For $K > 2$, the construction uses completely random experts (independent of y): - *World A*: $y = 0$, noise uniform over $K-1$ classes. Expert predicts 0 with prob $1-\epsilonpsilon_1$ - *World B*: y^* non-deterministic (same distribution as y in World A). Expert is pure random

$P_A(y=0|s1) = 1-\eta = P_B(y=0|s1)$ $P_A(f_m=0|s1) = 1-\epsilonpsilon_1 = P_B(f_m=0|s1)$ Independence of y and f_m holds in both worlds by construction. Joint distribution matches for all $K \geq 2$.

Error bound $\eta*\rho/2$

Claim: $\max(\text{Error_A}, \text{Error_B}) \geq \eta*\rho/2$

Verification: PASS.

Let a be the fraction of ambiguous samples ($\{x \text{ in } s1, y=1\}$) flagged as noise by the algorithm. Since observables are identical, a is the same in both

worlds. - World A: ambiguous samples are truly noise. Error contribution $\geq (1-a) * \eta * \rho$ - World B: ambiguous samples are truly clean. Error contribution $\geq a * \eta * \rho$ - $\max(\text{error_A}, \text{error_B}) \geq \eta \rho \max(1-a, a) \geq \eta \rho / 2$

The bound is tight: achieved at $a = 1/2$ (random guess on ambiguous set).

4. Theorem 4' (Exact Constant Minimax)

Bahadur-Rao saddlepoint

Claim: $\lambda^* = \log(\theta(1-p)/(p(1-\theta)))$, $\sigma^2(\theta) = \theta(1-\theta)$

Verification: PASS.

For $\text{Bern}(p)$: $\psi(\lambda) = \log(1-p + pe^\lambda)$ $\psi'(\lambda) = pe^\lambda / (1-p+pe^\lambda)$. Setting $= \theta$: $pe^\lambda = \theta(1-p+pe^\lambda) \Rightarrow e^\lambda = \theta(1-p)/(p(1-\theta)) \Rightarrow \lambda = \log(\theta(1-p)/(p(1-\theta)))$

Tilted variance: $\psi''(\lambda) = \theta(1-\theta)$. Verified by direct substitution: $1-p+pe^{\lambda^*} = (1-p)/(1-\theta)$, giving $\psi'' = \theta(1-\theta)$.

Chernoff information

Claim: $\kappa = KL(\theta_0 || \theta_1) = KL(\theta_1 || \theta_0)$ where $\theta^* = \log((1-p_0)/(1-p_1))/\log(p_1(1-p_0)/(p_0(1-p_1)))$

Verification: PASS.

The equality $KL(\theta||p_0) = KL(\theta||p_1)$ gives: $\theta \log(p_1/p_0) = (1-\theta)\log((1-p_0)/(1-p_1))$ Solving: $\theta^* = \log((1-p_0)/(1-p_1)) / \log(p_1(1-p_0)/(p_0(1-p_1)))$

Verified θ^* is in (p_0, p_1) by strict convexity of KL in its first argument and sign change at endpoints.

Adaptive threshold

Claim: $\theta_{\text{opt}} = \theta^* + (1/M)\log((1-\eta)/\eta)/D + O(1/M^2)$

Verification: PASS.

Minimizing $(1/2)\text{FNR}(\theta) + (1-\eta)/(2\eta)\text{FPR}(\theta)$ gives first-order condition: $KL(\theta||p_0) - KL(\theta||p_1) = (1/M)^*\log((1-\eta)/\eta)$

Key observation: $KL'(\theta||p) = 1/(\theta(1-\theta))$ is INDEPENDENT of p . Therefore: $KL(\theta+\delta||p_0) - KL(\theta+\delta||p_1) = \delta(KL'(\theta||p_0) - KL'(\theta||p_1)) + O(\delta^3) = \delta(\lambda_0^* - \lambda_1) + O(\delta^3) = \delta D^* + O(\delta^3)$

The δ^2 terms cancel exactly. This is a clean mathematical fact correctly exploited.

Therefore: $\delta = (1/M)\log((1-\eta)/\eta)/D + O(1/M^2)$.

$O(1)$ prefactor

Claim: $((1-\eta)/\eta)^s$ appears in both FPR and FNR contributions.

Verification: PASS.

At $\theta_{\text{opt}} = \theta^* + \delta$: $\exp(-MKL(\theta_{\text{opt}}/p_0)) = e^{\{-M\kappa\}}$
 $\cdot ((1-\eta)/\eta)^{\{-(1-s)\}} \exp(-MKL(\theta_{\text{opt}}/p_1)) = e^{\{-M\kappa\}} \cdot ((1-\eta)/\eta)^s$

where $s = |\lambda_1|/D$, $1-s = \lambda_0/D$.

FPR term: $((1-\eta)/(2\eta)) \exp(-M\kappa) ((1-\eta)/\eta)^{\{-(1-s)\}} / (\lambda_0 \sqrt{\dots})$
 $= \exp(-M\kappa) ((1-\eta)/\eta)^s / (2\lambda_0 \sqrt{\dots})$

FNR term: $(1/2) \exp(-M\kappa) ((1-\eta)/\eta)^s / (|\lambda_1| \sqrt{\dots})$

Both carry the identical factor $((1-\eta)/\eta)^s$. This cancellation is verified algebraically.

C_min formula

Claim: $C_{\min} = (\eta/2) ((1-\eta)/\eta)^s (1/\lambda_0 + 1/|\lambda_1|) / \sqrt{\theta(1-\theta)}$

Verification: PASS.

From the weighted risk at the Bayes optimal threshold: $R_M = w_0 \alpha_M + w_1 \beta_M = e^{\{-M\kappa\}} ((1-\eta)/\eta)^s (1/\lambda_0 + 1/|\lambda_1|) / (2\sqrt{2\pi M \theta(1-\theta)}) (1+o(1))$

$\lim e^{\{M\kappa\}} \sqrt{2\pi M} R_M = ((1-\eta)/\eta)^s (1/\lambda_0 + 1/|\lambda_1|) / (2\sqrt{\theta(1-\theta)}) = C_{\min}/\eta$

Solving: $C_{\min} = \eta \cdot [\text{above expression}] = (\eta/2) ((1-\eta)/\eta)^s (1/\lambda_0 + 1/|\lambda_1|) / \sqrt{\theta(1-\theta)}$

Constant matching

Claim: Adaptive SCX constant = $C_{\min}/\eta$

Verification: PASS.

From Lemma D section D.5: $1 - F_1(\theta_{\text{opt}}) \sim e^{\{-M\kappa\}((1-\eta)/\eta)} s(1/\lambda_0 + 1/|\lambda_1|)/(2s(\theta))$

This is exactly $C_{\min}/\eta$. The lower bound (Lemma E) proves no algorithm can have a smaller constant. SCX with adaptive threshold achieves it. Match is exact.

5. Theorem 5 (Cluster Consistency)

All exponents are negative

Claim: $-c_1 n_{\min} \Delta_{\min}^2 / (\sigma^2 d_{\phi})$ is always negative.

Verification: PASS.

The exponent contains: $c_1 > 0$ (universal constant), $n_{\min} > 0$ (minimum per-state sample size), $\Delta_{\min} > 0$ (minimum separation, fixed), $\sigma^2 < \infty$ (sub-Gaussian proxy), $d_{\phi} < \infty$ (feature dimension). All factors are positive, so the exponent is strictly negative. The bound converges to 0 as $n_{\min} \rightarrow \infty$.

Additionally verified: - Lemma S1: $P(\|\epsilon\|_2 \geq C_1 \sigma (\sqrt{d_{\phi}} + \sqrt{t})) \leq 2 \exp(-t)$. Exponent $-t$ is always negative for $t > 0$. - Lemma S2: Sub-Gaussian tail bound with negative exponent. - Lemma 2 peeling sum: each

term $\exp(-c4^j t^2)$ with negative exponent, summed exponentially.

All inequality directions

Claim: All inequalities point in correct directions.

Verification: PASS.

Verification table from Section 7.3 checked: - Lemma 1 Claim: $P(\text{event}) \leq \exp(-\Delta_{\min}^2/(8\sigma^2))$. The derivation: condition $\|\mu_j - \mu_k + \epsilon\| \leq \|\epsilon\|$ implies $2\epsilon T(\mu_j - \mu_k) \leq -\|\mu_j - \mu_k\|^2$. Sub-Gaussian tail gives the exp bound. Direction: upper bound (correct for tail). - Lemma 3: $\|\phi - \theta_{\{j\}}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{\{j'\}}\|$. Verifies $1/8 + 3/8 = 1/2$ and $7/8 - 3/8 = 1/2$. Both sides with correct inequality. - Lemma 2 (3): $\lambda/|\hat{\theta} - \theta|^2 \leq |(P_n - P)(f_{\hat{\theta}} - f_{\theta})|$. From $W_n(\hat{\theta}) \leq W_n(\theta)$, expanding and using quadratic lower bound. Direction: correct. - Lemma 4: $P(\text{no restart lands in } B) \leq (1-p_0)^R \leq n^{-c}$. Direction: correct.

NP-hard gap honestly discussed

Claim: The NP-hard gap of k-means is honestly acknowledged and addressed.

Verification: PASS.

Section 6 and 9.2 explicitly: 1. State that k-means is NP-hard in worst case 2. Lemma 4 proves Lloyd's with $R = O(\log n)$ random restarts finds global minimizer under strong separation 3. Explicitly state the consequence if strong separation fails: "if the strong separation condition does not hold...

the theorem’s guarantee degrades” 4. Cite relevant literature (Ostrovsky et al. 2013, Kumar & Kannan 2010) 5. Do not claim a guarantee for the weak separation regime

This is an honest, well-structured treatment of the computational hardness.

6. Numerical Verification

Test 1: $p_0=0.10$, $p_1=0.60$, $\eta=0.10$

Quantity	Computed Value	Verification
θ^*	0.31158	Closed-form formula
κ	0.1696	Matches Lemma C table (0.1696)
λ_0^*	1.4040	> 0 , correct direction
	λ_1^*	
D^*	2.6021	$\lambda_0^* +$
s	0.4604	
$C_{\min}$	0.4573	Formula evaluated
$C_{\min}/\eta$	4.573	
C_{SCX}	4.573	$C_{\text{SCX}} = C_{\min}/\eta$: MATCH

$\kappa/2\Delta^2$ comparison: $2(p_1-p_0)^2 = 20.25 = 0.50$. $\kappa/0.50 = 0.1696/0.50 = 0.3392$, matching Lemma C table.

Test 2: $p_0=0.05$, $p_1=0.80$, $\eta=0.05$

Quantity	Computed Value	Verification
theta*	0.35979	Closed-form formula
kappa	0.4576	Matches Lemma C table (0.4574, minor rounding)
lambda0*	2.368	> 0
	lambda1*	
D*	4.330	lambda0* +
s	0.4531	
C_min	0.1863	Formula evaluated
C_min/eta	3.727	
C_SCX	3.727	C_SCX = C_min/eta: MATCH

Test 3: p0=0.20, p1=0.50, eta=0.30

Quantity	Computed Value	Verification
theta*	0.33904	Closed-form formula
kappa	0.0527	Matches Lemma C table (0.0528, minor rounding)
lambda0*	0.7186	> 0
	lambda1*	
D*	1.3859	lambda0* +
s	0.4815	
C_min	1.373	Formula evaluated
C_min/eta	4.578	
C_SCX	4.578	C_SCX = C_min/eta: MATCH

All three numerical test cases confirm the identity $C_SCX = C_min/eta$.

7. Minor Observations and Notes

1. **Lemma A lattice correction:** The Bahadur-Rao expansion for Bernoulli distributions includes a lattice correction factor $(1 - e^{\{-\lambda\}})^{-1}$ *not present in the continuous case*. The document correctly notes both the $1/\lambda$ form and the lattice correction form, showing they are asymptotically equivalent. When used consistently in FPR/FNR ratios (which share the same θ^*), the correction largely cancels. This is mathematically sound.
 2. **Theorem 2 Lipschitz constant C_F :** The Lipschitz constant for F1 as a function of (TP, FP, FN) depends on the operating regime. The document gives heuristic values ($C_F \leq 3$ for precision, recall ≥ 0.1 ; $C_F \leq 1$ for precision, recall ≥ 0.5). These are not rigorous bounds derived in the document, but the claim that a Lipschitz constant exists is correct. The specific values are practical estimates.
 3. **Theorem 4' non-asymptotic bounds:** Section 6(d) claims finite-M bounds with explicit constants $K1$, $K2$, but states these come from Lemma A (Stirling error) and Lemma B (quadratic remainder). The lemmas provide error terms of the form $O(1/M)$, but the explicit $K1$, $K2$ constants are referenced rather than fully computed. The asymptotic limit ($M \rightarrow \infty$) is fully rigorous.
 4. **Theorem 5 irreducible error:** The document acknowledges that even with perfect center estimates, points with $\|\epsilon\| \geq 3\Delta_{\min}/8$ can be misclassified. This is correctly identified as a data property, not an estimator failure.
-

8. Overall Mathematical Verdict

All mathematical derivations across Theorems 1-5 and supporting lemmas are **CORRECT**.

Theorem	Verdict	Key Findings
Theorem 1 (Noise Detection)	PASS	All lemmas algebraically verified; Hoeffding conditions satisfied; F1 bound derivations correct
Theorem 2 (Weak Features)	PASS	Fano/Pinsker/DPI applied correctly; TV-to-F1 Lipschitz argument valid (constants are heuristic but not essential)
Theorem 3 (Unidentifiability)	PASS	K=2 and K>2 constructions both verified; joint distribution equality holds; error bound $\eta \cdot \rho / 2$ is tight
Theorem 4' (Exact Constant)	PASS	Bahadur-Rao verified in full; Chernoff θ^* closed-form correct; adaptive threshold derivation exploits KL' independence of p; C_min formula matches all three numerical tests; C_SCX = C_min/ η verified

Theorem	Verdict	Key Findings
Theorem 5 (Cluster Consistency)	PASS	All exponents negative; all inequality directions verified; NP-hard gap honestly discussed and addressed under strong separation
Lemma A (Bahadur-Rao)	PASS	Full saddlepoint computation verified; tilted distribution correctly identified as Bern(theta); lattice correction discussed
Lemma B (F1 expansion)	PASS	F1 formula algebra verified; denominator positivity checked; expansion into $\text{FNR}/2 + (1-\text{eta})\text{FPR}/(2\text{eta})$ correct; remainder bound derived
Lemma C (Chernoff info)	PASS	Theta* closed-form derived correctly; KL'' independence noted; numerical table entries verified against formulas
Lemma D (Adaptive threshold)	PASS	First-order condition derived correctly; $O(1/M)$ shift computed; $(1-\text{eta})/\text{eta}$ cancellation verified

Theorem	Verdict	Key Findings
Lemma E (Lower bound)	PASS	Bayes test reduction via Neyman-Pearson is correct; Bahadur-Rao applied; $C_{\min}$ formula derived explicitly and matches achievable constant
Lemma F (Multi-state)	PASS	Additivity by law of total expectation; bottleneck state dominates asymptotic; correct handling of $\kappa_{\min}$ states